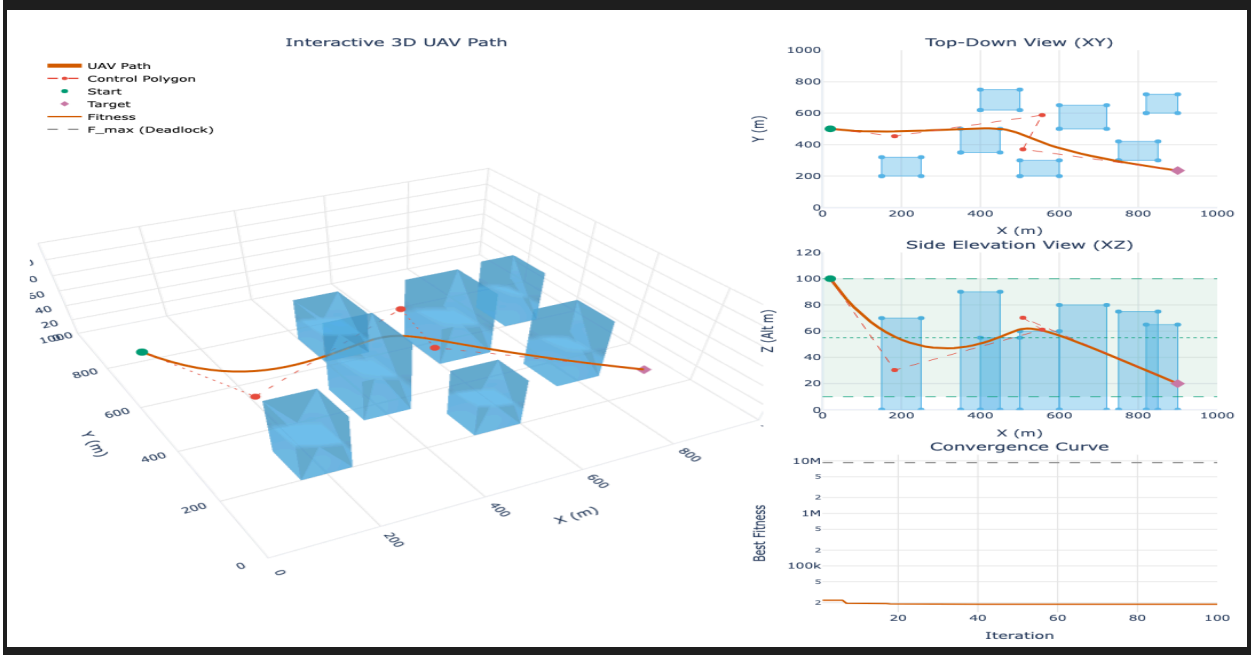
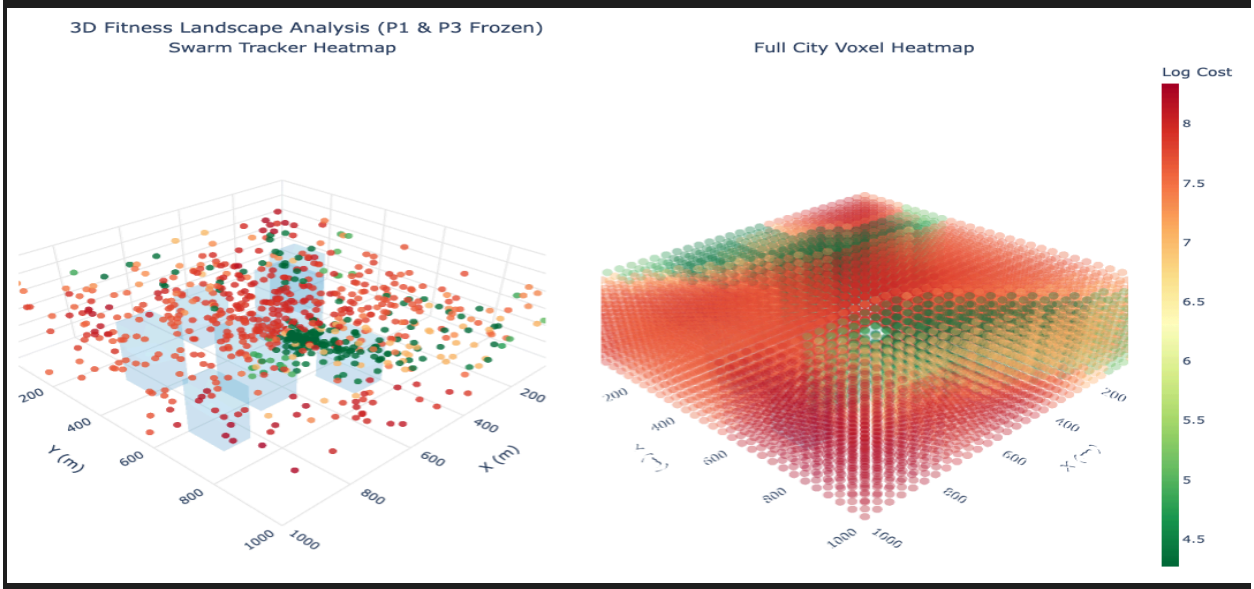
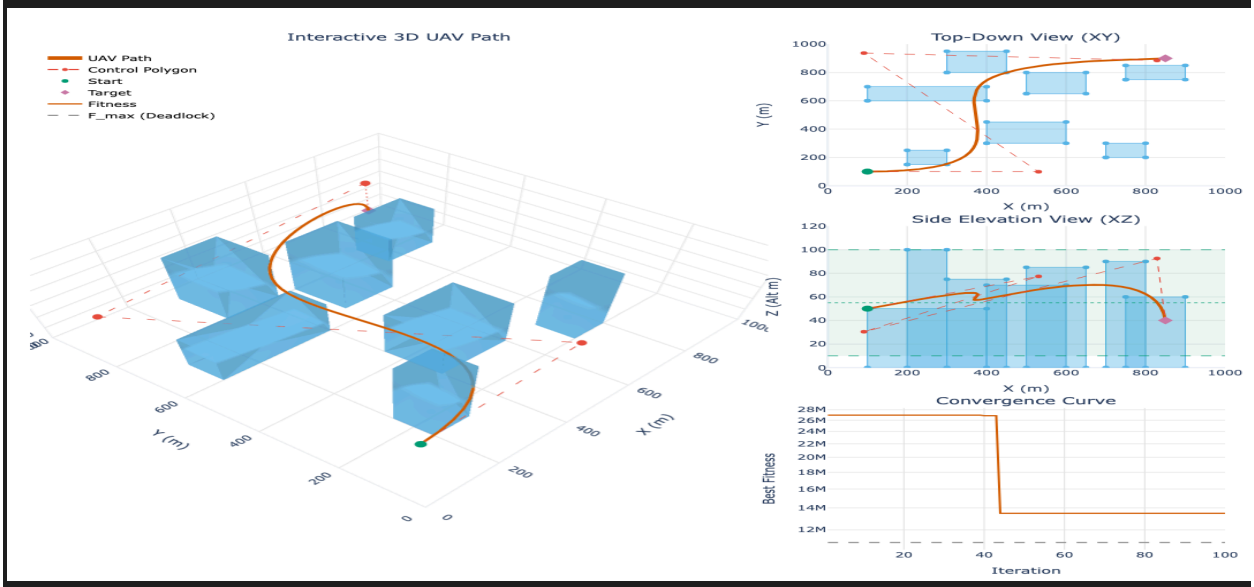
```
1)    START  = np.array([20.0,  500.0, 100.0])
      TARGET = np.array([900.0, 235.0, 20.0])
      OBSTACLES =    [[150, 250, 200, 320,  0,  70],
                      [350, 450, 350, 500,  0,  90],
                      [500, 600, 200, 300,  0,  60],
                      [600, 720, 500, 650,  0,  80],
                      [750, 850, 300, 420,  0,  75],
                      [820, 900, 600, 720,  0,  65],
                      [400, 500, 620, 750,  0,  55],]
```

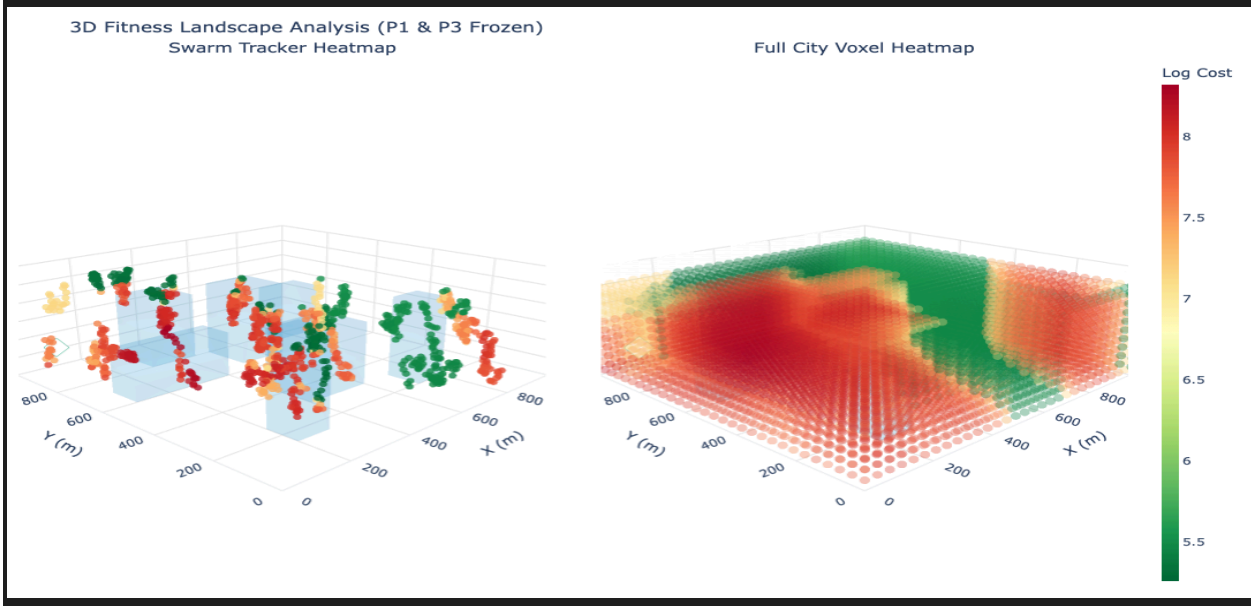




TEST-CASE 2:

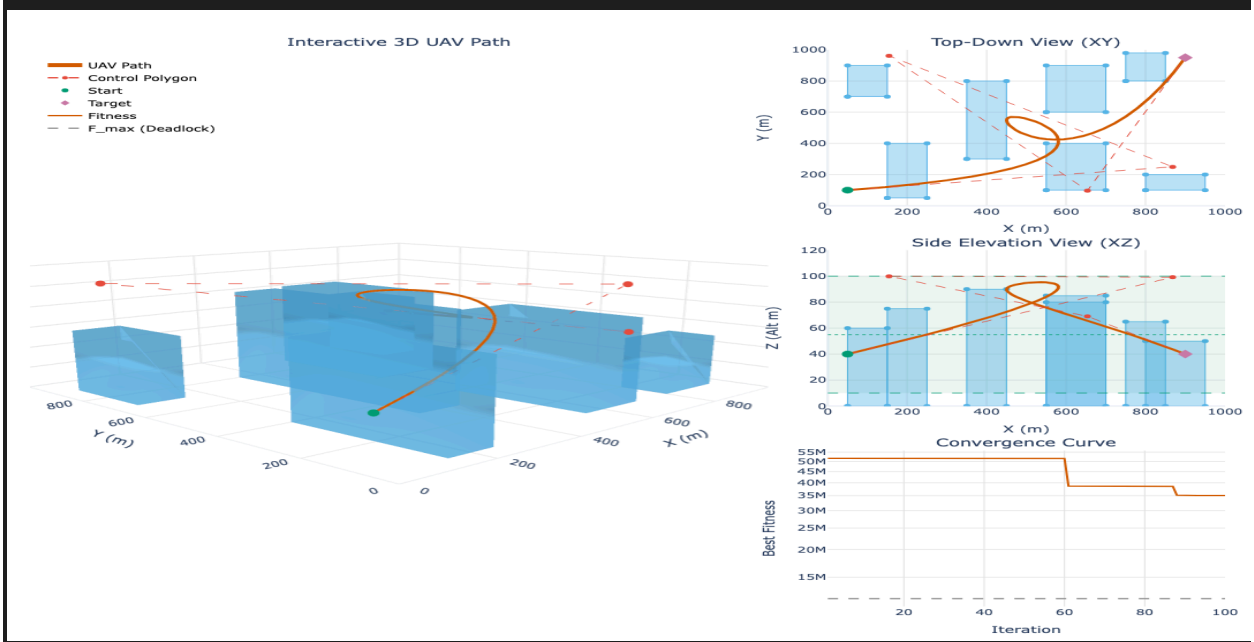
```
START  = np.array([100.0,  100.0,  50.0])
TARGET = np.array([850.0,  900.0,  40.0])
OBSTACLES =    [[200, 300, 150, 250, 0, 100],
               [400, 600, 300, 450, 0,  70],
               [700, 800, 200, 300, 0,  90],
               [100, 400, 600, 700, 0,  50],
               [500, 650, 650, 800, 0,  85],
               [750, 900, 750, 850, 0,  60],
               [300, 450, 800, 950, 0,  75],]
```

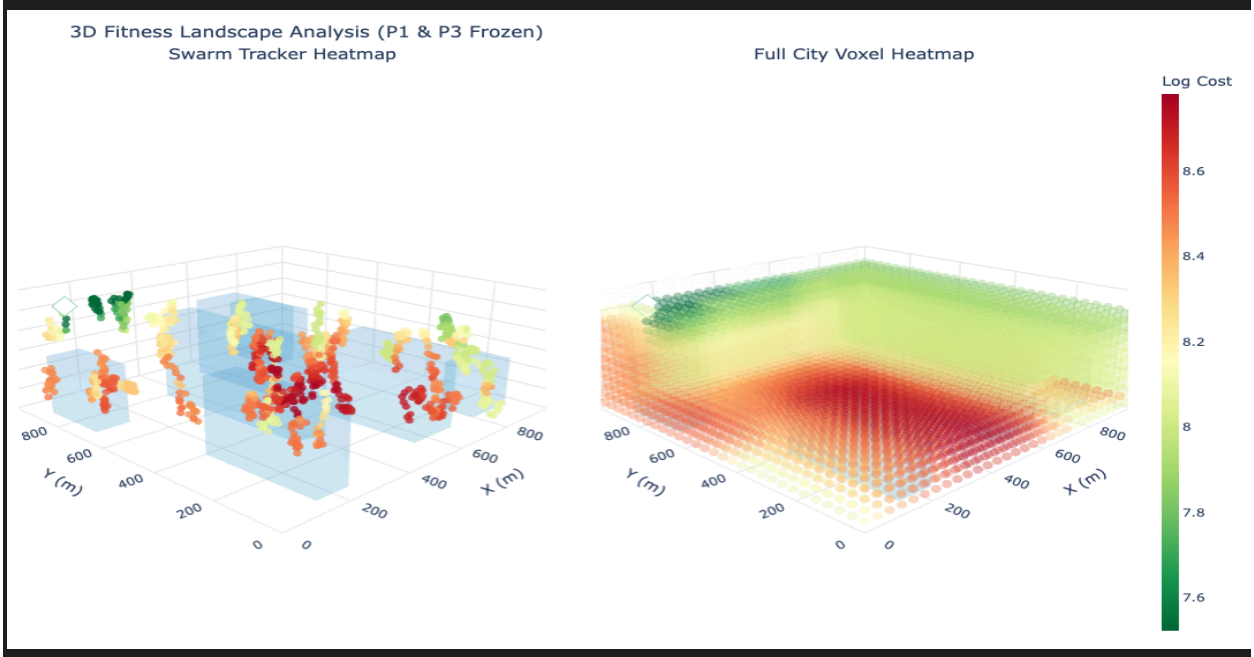




TEST-CASE 3:

```
START  = np.array([50.0, 100.0, 40.0])
TARGET = np.array([900.0, 950.0, 40.0])
OBSTACLES = [
    [150, 250, 50, 400, 0, 75], [350, 450, 300, 800, 0, 90],
    [550, 700, 100, 400, 0, 80], [550, 700, 600, 900, 0, 85],
    [750, 850, 800, 980, 0, 65], [800, 950, 100, 200, 0, 50],
    [50, 150, 700, 900, 0, 60]]
```





Visualizing Algorithm Dynamics: Swarm Tracker Heatmap Interpretation

1. Global Exploration: The "Scattered" Swarm (Test Case 1)

When the 3D data points are spread widely across the entire bounding box, resembling a scattered cloud, the algorithm is in its global exploration phase.

- **Algorithm Phase:** Occurs during the early iterations (typically t = 1 to 30).
- **Interpretation:** The inertia weight (w) is at its maximum. Particles ignore neighborhood influences and maintain high velocities to aggressively map the entire 3D voxel space.
- **Evaluation:** A scattered state is essential initially as it proves the algorithm is not prematurely converging. However, if the swarm remains scattered by iteration 90, it indicates a failure to locate a safe flight corridor.

2. Local Exploitation: The "Clustered" Swarm (Demonstrated in Test Cases 2 & 3)

When the data points are tightly packed into a dense sphere or focused line, the algorithm has transitioned into local exploitation.

- **Algorithm Phase:** Occurs during the later iterations (typically t = 60 to 100).
- **Mathematical Interpretation:** The inertia weight (w) has decayed, allowing the social learning factor (c_2) to dominate. The swarm has reached consensus on the 3D coordinate offering the lowest penalty and is now micro-adjusting positions to minimize the final distance cost (f_L).
- **Visual Indicators & States:**
    - **The Convergence State (Green):** The cluster is tightly packed in a green zone. This proves the swarm successfully identified the optimal gap, threaded the spatial constraints, and achieved global convergence.

The MSIPSO Deadlock Resolution ("The Jump")

A critical feature of the MSIPSO architecture is its deadlock escape mechanism. If a tight cluster suddenly dissolves and reverts to a scattered cloud in the subsequent iteration, the algorithm has successfully detected a Trap State. Recognizing that the cluster is trapped in an invalid penalty zone (> F_{max}), the algorithm deletes current velocities and redistributes the swarm via random affine transformations to force new exploration.